Abstract

We record a complete research trail — from a transformer training observation (2026-07-29) through a systematic experiment series (E1–E18), the extraction of structural laws, the construction of the Pat framework (basepoint-relative operator semantics), and its full Lean 4 + mathlib formalization (claims R136–R153 and beyond; all PROVED, no `sorry`; full `lake build` passes, 3631 jobs). The trail is anchored by timestamped evidence: session records, a 2303-turn task log, git history, and experiment artifacts. The starting observation is the 0/1 OOD dichotomy: a low-epoch (60) transformer reaches training accuracy 1.000 yet generalizes 0/32 on a same-basepoint notation change, and 32/32 after equivalence-declaration bridging — a reproducible two-pole pattern across 45 training runs. From this observation the trail extracts five structural laws (anchor binding, notation translatability, shift invariance, structure-bearing representation, definition-layer invisibility), maps them to framework claims (R136 direction declaration, R138 phase locking, R143 interlock matrix), and formalizes the chain to the continuum construction (R150, R151, R163). We contribute (i) the complete experiment-to-formalization path as a reproducible artifact — each experiment documented individually with design reason, procedure, result, and significance — and (ii) a methodology for recording research trails under provenance pressure: SHA-256-anchored manifests, per-claim publication, and a self-contained reproducibility package.

Lean, formalization, transformer, OOD, basepoint, reproducibility, research trail

An Empirical Trail to Formalized Pat Theory: from 0/1 OOD Observations to Machine-Checked Theorems (complete research path, timestamped evidence, all PROVED, no `sorry`)

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2026-08-20

1 Introduction

This paper is a complete, machine-checkable record of a research trail: how a transformer training observation became a set of structural laws, then a framework, then Lean-formalized theorems. We record the trail because it is itself the contribution: the path from an empirical two-pole OOD pattern to a formalized theory of basepoint-relative operator semantics, with every step timestamped and reproducible.

The trail starts on 2026-07-29 with a first *imperfect* generalization observation (relation-leave-one-out generalization, 22:39:12 UTC+8) — imperfect, yet already sufficient to guide the exploration direction. It reaches its first *perfect* generalization on 2026-08-15 04:01 (all 19 operations OOD 0.000 $\to$ 1.000). The core two-pole pattern (E12) is: a 60-epoch, 2-layer transformer (dim 64) reaches training accuracy 1.000, yet scores $\mathbf{0/32}$ on a same-basepoint notation change ($\mathrm{succ}_a \to \mathrm{succ}_b$) and $\mathbf{32/32}$ after adding equivalence-declaration bridging samples — reproduced across 45 training runs (5 groups $\times$ 3 seeds $\times$ 3 independent runs), with no deviation.

From this pattern, a systematic experiment series (E1–E18) extracts five structural laws; the laws map to the Pat framework (basepoint-relative operators: every operator has its own basepoint, directions must be declared in pairs); the framework is formalized in Lean 4 (mathlib v4.32.2), claims R136–R153 all PROVED with no `sorry` (completed 2026-08-13 13:31), and the chain extends to the continuum construction (R150 Wang's phase-locking theorem, R151 continuum as Pat-grid closure, R163 endogenous construction with zero Nat). Full `lake build`: 3631 jobs, no `sorry`.

Contributions.

1. The complete empirical trail E1–E18, each experiment documented individually (§§3.1–3.18): design reason, procedure, result, and significance, as a reproducible artifact (20 scripts, 15 result files, one-command reproduction), with early first-successes recovered from the session log (2026-07-29/07-30 prefigurations; perfect generalization 2026-08-15 04:01; reproduction 2026-08-16 14:54, per-seed identical).

2. The Riemann direction (C011–C025, 2026-08-06 visualization → 2026-08-12 formalization and publication) and the homework-exercise series (2026-08-13 → 08-15, 25 reports, each with Lean 0-`sorry`).

3. The five structural laws extracted from the observations, and their mapping to formalized framework claims (R136/R138/R143/R150/R151/R163).

4. The logical completeness theorem (decoupling operator D = R1 exclusion + R2 cancellation + R3 layering $\Longrightarrow$ completeness at any order/element/subject-object, verified 2026-08-15 09:14–09:47 with the two-symmetry-groups corollary I7v) and the three mapping-judgment theorems (symbol-norm / presentation-legality / intuition-precision, formalized in `MappingJudgmentTheorems.lean`, 0 `sorry`, 2026-08-16).

5. The Lean formalization (150+ Toolkit files, claims R001–R232, 0 `sorry`, 3631 jobs).

6. A timestamped evidence chain (session records, 2303-turn task log, git history, file timestamps) and a methodology for recording research trails under provenance pressure.

2 The starting observation: the 0/1 OOD dichotomy

2.1 Timeline anchors

- **2026-07-29** (session start): first *imperfect* generalization. A pure next-token transformer memorizes 2-digit addition 100% (3-digit 0%). At 22:39:12 (UTC+8) the relation-leave-one-out generalization is confirmed (training 5 relations, leaving 1: ≤ generalization 81% → 90% with a new architecture). This is the trail's first generalization observation. It is *imperfect* — not all operations generalize — but it is already directional: it shows that generalization is possible without the "necessary number of steps", and it determines the exploration direction (which representations generalize, which collapse). The same evening produced the first per-token analysis (22:41, revealing the model's "reverse" strategy — the origin of the token-level reporting discipline) and the four-question-type system (judgment / choice / fill-in / computation, 22:49).

- **2026-07-30**: early first-successes that prefigure the E-series. Cross-class generalization (arithmetic/relations $\rightarrow$ logic) 90% same-class / 42.3% cross-class (00:03); logic-operator leave-one-out (leave $\equiv$ 90%, leave $\rightarrow$ 92%, leave $\leftrightarrow$ 82%, 00:28); base-change equivalence $\rightarrow$ addition generalization 81% (00:42); the judgment paradigm (soft 84% / vq 71%) and the parallel-token verification (leave $\leq$ breaking the counterpart axis: 55% vs 90% with complete counterparts, 05:30); the $\leq$ generalization 47% $\rightarrow$ 79% when logic-gate co-training completed the parallel counterparts (08:45). These are the first instantiations of what E-series re-runs later formalized as E1/E3/E7–E13.
- **2026-08-07**: the basepoint-movement theory is proposed by the user (thinking document *On the Riemann Conjecture*: the lambda origin drifts on the circle of radius 1 around 1; iteration continues along the drift path; the complex plane's origin is suspect — the two axes' origins are by definition very likely completely different). The `relative-recursion` project starts the same morning (10:30) with the basepoint question as its working problem, and its first Lean no-`sorry` (C001 semiconjugacy) lands at 11:31.
- **2026-08-09 01:27**: the user's base(base,base) concept (ChatGPT conversation "Lambda calculus and Church numerals") — the lambda basepoint self-application presupposes an operation concept distinct from the basepoint (lambda(base, base)), while the intended concept has none (base(base, base)). Same day it becomes the `pure-relative` branch of the Lean formalization (claim P001 PureBase; `CONoGo`, `SelfRef`, `MinimalRuleAudit`).
- **2026-08-15 04:01**: first *perfect* generalization: all 19 operations learned, OOD 0.000 $\rightarrow$ 1.000 per epoch (reaching 1.000 at epoch 14). The E-series re-runs in the v5 token system follow at 06:48–09:00 (experiment day E1–E13).
- **2026-08-16 14:54**: the user asked to reproduce the transformer experiments; E12 was reproduced 15 training runs per-seed identical, plus a third in-package run (self-contained package, 08-16 15:04). E14–E18 followed the same day.

2.2 The daily record, 2026-08-01 $\rightarrow$ 08-07

Every day in this window carried work — trainer design, token-system theory, tooling, and the launch of the basepoint-movement research. The window has one unifying theme, stated by the user on 08-02 and verified by every subsequent day: *can the model's trained intuition be induced, rather than corrected?* The user's principle: "what you should do is make the default methods that are trained into the model succeed" — the infrastructure must route the model's native intuitions, not replace them with new behavior. The R-series prompt experiments (08-03) are the direct test: pseudo-code headers, code anchors, and few-shot templates each *induce* better execution (R8 97 vs R11 30 without them), while negative prohibitions and over-constraint are shown to *mislead* the intuition (prohibitions suggest the method; tight constraints collapse). The 08-07 trigger-rate work (0% $\to$ 40.9%) tests the precondition of induction — the skill must fire before it can steer; and C001's reuse of mathlib's `Semiconj.iterate_right` instead of re-deriving is the induction of the model's retrieval intuition. The same theme continues after this window: the CoT-before tool hint (lean-cot-assist, 08-07/08-09) that lowers the 27B's algebraic expansion load, and the R-series re-runs on the local 27B after 08-14. The record is explicit about what was verified: induction works (prompt shape measurably steers execution), induction has limits (negative instructions invert), and induction is toolable (trigger description, hijack-to-correct, hint injection).

The day-by-day record (from the session log, 08-01: 11 entries, 08-02: 31, 08-03: 17, 08-04: 8, 08-05: 13, 08-06: 5, 08-07: 21): The day-by-day record (from the session log, 08-01: 11 entries, 08-02: 31, 08-03: 17, 08-04: 8, 08-05: 13, 08-06: 5, 08-07: 21):

08-01 — trainer deepening and token completeness (11 entries). Generalization supervision (train/val split) landed; the user corrected the input format: "you are wrong — the input must be an element sequence, where every element is composed of its own logic tokens" (the raw-string input was rejected). A triple-search harness (three parallel agents over module-A $\times$ module-B $\times$ router) found the best combination (layer=tensor, transit=overlap, router=scalar_only); full triple enumeration gave 39.8% overall generalization. The strict $a + b$ extrapolation scored $0/52 = 0\%$ — the user's diagnosis: "parallel element / parallel token training is insufficient; from the start we must train addition/subtraction/multiplication, bit operations, base changes, powers, all logic operations, all question types, with the best method together". The user's three completeness questions fixed the token design: (i) is there a question type for every token's generalization? (ii) should value not be a scalar? (iii) is every derive traced to baseloop with training samples designed? — resulting in 25 zero-coverage token question types, value dtype scalar $\to$ int/classification, and definition propositions 16 $\to$ 46 (42/42 complete). The meta-layer was acknowledged: let the model pour information into itself during training.

08-02 — tooling infrastructure and the intuition-adaptation beginning (31 entries). Llama-server management skill; diagnostic-tool plugin system (12 plugins); the technology choice ".sh only for light orchestration, resident services in JS" (sh-infrastructure skill). The user's central correction launched the trail-long intuition-adaptation methodology: "what you should do is make the default methods that are trained into qwythos succeed" (instead of correcting the model's behavior) — the principle that infrastructure must route the model's native intuitions, not replace them. Prefix-cache-hit analysis and statusline data integration into claude local; meta-logic/meta-method/ meta-rule research (complete meta-concept landscape, 8 files); the architecture workflow "research $\to$ design $\to$ implement" with fact-check and judgment-check at every step.

08-03 — skill writing method and prompt iteration (17 entries). The user's corrections became rules: pseudo-code must use code commands ("run fn", not "execute"); negative prohibitions act as hints ("prohibitions directly violate our experience; they don't prevent, they suggest the method") — rules must be positively phrased. The R-series prompt experiments (R8: 97 requirement lines with pseudo-code header + code anchors; ablation R9=46 / R10=63 / R11=30; R12=57 / R13=72 with few-shot templates) established the prompt anatomy: natural-language task name first (token direction), then pseudo-code activation, few-shot templates as format anchors, no over-constraint. Self-generated prompt research: have the 27B iterate a bad prompt into R13 quality (the evaluation script was rejected as answer leakage). Deterministic errors are treated as 100% errors: hijack the wrong command to the correct one, or instruct the model which alternative method to use.

08-04 — technology-stack switch and architecture workflow (8 entries). opencode installation failure diagnosis (CachyOS mirror 404s); the switch to opencode Go via a ccproxy (Rust bridge cut from cc-switch, anthropic-to-openai mode, dual providers); statusline migrated verbatim from `.claude` ("I didn't ask you to change the color scheme or the logic"); the dsv4-flash 1M context displayed as 200k fixed at the gateway (n_ctx 1000000); skill-activation research led to skill-matcher.js (automatic text matching, no AI model in the loop); G-layer/SKI error summary document.

08-05 — skill desensitization and trigger-rate optimization (13 entries). Switch to opencode dsv4 with the new ccproxy (Rust bridge with first-byte timeout failover); token-system skill improved; steering-8b local-run exploration (causal diffusion LM — not supported by llama.cpp/GGUF; transformers + bitsandbytes 4-bit path prepared); the harness results published to GitHub (harness-for-all) after the user deleted the first 37-file release and required only two skills, fully desensitized (no claude traces, no dynamic-workflow traces); all 7 skills desensitized; RTX PRO 5000 evaluation for DeepSeek-V4-Flash-MTP-DSpark-MLX (conclusion: cannot run — MLX is Apple-only and the model is a 304B MoE).

08-06 — bridge startup and the Riemann visualization (5 entries). bridge/start.sh + opencode.sh optimization (stale-socket cleanup, flock inheritance fix); the "170hx unlocks 80G VRAM" claim verified (true mechanism: CMP 170HX is a full GA100 die; stable community range 10G → 40G); and the Riemann zeta / prime-position visualization project (17:10 request → 17:55 delivery): 52/52 numerical verifications, Lehmer $\pi(x)$ to 10^{13}, browser-tested interactions, complex-base (Gaussian-integer) support with 14 valid bases (2800 round-trip tests pass), and persisted prime/zero data (664,579 primes; 138,067 zeros; Riemann–Siegel correction 0.1 → 9×10^{-4}).

08-07 — basepoint-movement research launch + first Lean no-`sorry` (21 entries). The `relative-recursion` project started (10:30) with the basepoint question as its working problem (the user's thinking document *On the Riemann*

Conjecture: basepoint drift, iteration along the drift path, the suspect origin); the C001 semiconjugacy formalization completed with no `sorry` at 11:31 (first Lean commit 10:53, reusing mathlib's `Semiconj.iterate_right`); the toolchain was built the same day (allowlist plugin system with hot reload and masking; websearch with –author and on-disk results; the natural-number-origin research discipline written into the repository rules; T1–T3 verification tools; info_units four-layer information elements; the discipline-guard hook; the Mathematical Research Protocol skill; description trigger-rate diagnosis 0% → 40.9%).

2.3 The two-pole pattern (E12)

Group	Training samples	OOD target	OOD (45 runs)
1a baseline	succ_a 250	succ_a	$\mathbf{32/32} \times 9$
1b zero-shot	succ_a 250	$\mathbf{succ}_b$	$\mathbf{0/32} \times 9$
2 bridging	succ_a + equivalence 270	$\mathbf{succ}_b$	$\mathbf{32/32} \times 9$
$2'$ control	succ_a + equivalence 270	succ_a	$32/32 \times 9$
4 symmetry	$\text{succ}_a/\text{succ}_b$ mixed 252	mixed	$32/32 \times 9$

All 45 runs reach training accuracy 1.000; the OOD two-pole (0/32 vs 32/32) never deviates. Token-level analysis locates the collapse: the 0-pole fails at the notation-branching position (input head passes 32/32; the failure is the notation fork), a mechanism fingerprint invisible to aggregate accuracy.

Interpretation: training intuition is notation-bound statistics; a 1.000 training accuracy does not transfer to an untrained notation; OOD 1.0 only shows that *statistics extrapolate within that representation.* The equivalence-declaration samples (pairwise declarations, anchoring errors irrelevant) bridge the two notations — declarations make the information point at the correct structure.

3 The Riemann direction and the basepoint-movement theory (2026-08-06 → 08-12, before the experiments)

Before the experiment series, the trail ran a parallel formalization branch: the Riemann direction (C011–C025). Its intellectual spine is the basepoint-movement theory, proposed by the user on 2026-08-07 (the thinking document and the `relative-recursion` project session), formalized in Lean in the same project the same day (C001, 11:31), carried into the Riemann direction on 2026-08-12 (basepoint drift under projection, C011), and finally witnessed empirically by the training experiments (E11 anchor binding, 2026-08-15 07:43). It starts from the visualization project (2026-08-06 17:10 request → 17:55 delivery, within 45 minutes: the Riemann zeta landscape, the critical line $\sigma = 1/2$ highlighted with nontrivial zeros plotted, prime positions $\leq 10^8$, $\pi(x)$ approximations $\leq 10^{13}$, bases 2–36 and Gaussian-integer complex bases; the complex-base extension and the critical-line-only layout followed at 17:59–18:09, and the persisted prime/zero data at 18:35–18:46, 138,067 zeros with a Riemann–Siegel correction fix bringing the zero-location error from 0.1 down to 9×10^{-4}). The user's thinking document *On the Riemann Conjecture* (written around 2026-08-07, between the visualization and the formalization) records the intuition behind the direction: primes have no construction method (all other numbers have many); irrationals are the remainder of a higher-dimensional structure after projection lost part of its carrying structure; and the pair 1 vs $1/a + 1/b$ (translation vs inversion) leads to the **basepoint-drift conjecture — the origin of the basepoint-movement theory, 2026-08-07**: the lambda-calculus origin is not fixed; it moves on the circle of radius 1 around 1, and when the drifted basepoint's successor iteration is projected back to our number line, the projection lands on primes. The document adds a second hypothesis: if the basepoint drifts, iteration continues *along the drift path* — then primes land exactly on the translated integer positions, and one can observe the integer points on that axis. The basepoint-movement research itself had started the same day in the `relative-recursion` project (session 10:30; the project's working question: structures without a privileged zero, choosing different basepoints interpreted as "0") with its first Lean no-`sorry` at 11:31 (C001 semiconjugacy); the thinking document refers to this as "the basepoint-drift experiment Lean formalization I did earlier".

On 2026-08-09 01:27:22 (UTC+8) the user's earliest basepoint-formalization concept landed in a shared ChatGPT conversation ("Lambda calculus and Church numerals", archived in the artifact): the lambda-calculus basepoint self-application iteration *presupposes a concept distinct from the basepoint* — the self-application operation itself, written lambda(base, base); but in the intended experimental concept there is no such operation, written base(base, base) — the same primitive occupies the operator, operand, and basepoint positions. This base(base, base) design became the `pure-relative` branch of the formalization the same day (claim P001, `PureBase`; Lean files `CONoGo`, `SelfRef`, `MinimalRuleAudit`), the earliest Lean embodiment of the basepoint-movement idea: the basepoint is not a distinguished object but a role. The suspicion about the complex plane's origin belongs to the same period and the same direction: the two axes' origins, by their definitions, are very likely *completely different* — the origin is not trustworthy. In the 08-12 session the user makes this explicit and, from then on in the session log, refers to the complex plane's real axis as the "fake real axis" (*jia shi shu zhou*).

The Lean formalization session ran 2026-08-12 00:00–02:54:

- 00:00–00:12: six prime-drift position observations formalized (`PrimeDriftPositions.lean`, no `sorry`);
- 00:16–00:21: the user's projection hypothesis formalized as the true complex axis — a two-dimensional rotation algebra $a + bJ$, $J^2 = -1$, where $\sqrt{-1}$ exists *before* projection (`ComplexAxis.lean`);
- 00:23–00:27: the user re-states the 08-07 basepoint-drift conjecture against

the freshly built complex axis and sharpens it: since the complex plane's axis carries the naturals, the lambda basepoint has moved onto the complex axis — *the basepoint has drifted, its position is not the origin but some transform of* i (user, 00:23, quoted in the session log). Then (00:25): "that so-called real axis is very suspicious — I suspect it is not a real number axis at all, but another fake complex axis". This is where the "fake real axis" term enters the log; the user uses it consistently from then on. Both statements were formalized the same night: the basepoint drift (the true position is unobservable under projection) and the fake-real-axis theorem (the real axis is a sub-algebra $\{\langle a, 0\rangle\}$ of the same structure, its position unobservable — the "origin illusion");

- 00:28–00:29: C011 + C012 filed as the first claims of the direction;
- 00:53–02:04: the chain C013–C025 (projection frame, prime circles C012–C018, critical-line geometry C019–C022, prime-circle products C023–C024, Euler-product convergence with the zero-free region C025);
- 02:54: session archived with C011–C022.

The claim chain C011–C025, item by item. The direction is a single intuition chain, each link formalized in Lean (all PROVED, novelty KNOWN — restatements of classical mathematics; the record states explicitly, per the user's instruction, that no claim of proving the Riemann hypothesis is made and mathlib's official statement remains unproved):

1. **C011 (projection construction of the complex plane)**: the complex axis is the projection of a two-dimensional rotation algebra $ComplexAxis = \{\langle a, b\rangle\}$, $J = \langle 0, 1\rangle$, $J^2 = -1$; $\sqrt{-1}$ exists before projection; basepoint drift under projection (the basepoint's true position is unobservable — the user's "fake real axis" is the theorem: the real axis is a sub-algebra of the complex rotation algebra, not an independent axis; its position is unobservable under projection — the "origin illusion"). Filed 08-12 00:29 (commit 133f353).
2. **C012 (prime drift positions)**: primes land on translated integer points of the drift axis; six positional observations (integer decomposition $n \cdot p$ prime $\Rightarrow n = 1$, irrationals have no integer multiple, harmonic identities) formalized in `PrimeDriftPositions.lean`, 00:00–00:12, commit 7fdcb91.
3. **C013 (Riemann direction in the projection frame)**: inversion drift, partial sums, and the critical line as 15 theorems in the projection frame, 00:53–01:02 (commit 7c0feb8).
4. **C014 (sums of two squares, C015 curling)**: a prime $p \not\equiv 3 \pmod 4$ is the norm $a^2 + b^2 = p$ of some complex point (citing mathlib's Fermat two-squares theorem); the inversion $\operatorname{recip} z = \overline{z}/|z|^2$ curls infinity back to finiteness (for every ε there is R with $|z| > R \Rightarrow |\operatorname{recip} z| < \varepsilon$), is an

involution, and satisfies $|\text{recip}\, z| = 1/|z|$ — the geometric mechanism of analytic continuation. C014 commit 20df6b6 (01:13–01:21), C015 commit de3e966 (01:38).

5. **C016 (lattice points on a circle)**: the orbit structure of integer points on prime circles, uniqueness via the Gaussian-integer UFD ($\mathbb{Z}[i]$ is a Euclidean domain), 01:42 (commit e235d09).
6. **C017 (prime circles and conjugate pairing)**: a prime $p \equiv 1 \pmod 4$ circle carries exactly one orbit of 8 lattice points (up to units $\pm 1, \pm i$), splitting into conjugate Gaussian primes; 01:55 (commit edbc099).
7. **C018–C022 (critical-line geometry)**: the critical line as the inversion–translation duality center (C019), the critical-line circle through 0 and 2 (C020), the reflection-condition equivalence (line $\iff$ reflection condition $\iff$ circle), and the functional- equation symmetry restated algebraically — explicitly not a claim about zero positions.
8. **C023–C024 (prime-circle products and geometric splitting)**: products of prime circles and their decomposition into conjugate pairs under rotation.
9. **C025 (Euler-product convergence and zero-free region)**: for $\text{Re}(s) > 1$, $\prod_p (1-p^{-s})^{-1} = \sum_n 1/n^s =$ mathlib's official $\zeta(s)$ (via `eulerProduct_completelyMultiplicative` + `Complex.summable_one_div_nat_cpow`); $\text{Re} \geq 1$ is zero-free.

The honest boundary is recorded in the claim chain itself: every claim is labeled KNOWN (restatement), the Riemann hypothesis and the twin-prime conjecture are neither proved nor touched, and the positional claim on zeros (all nontrivial zeros on $\text{Re} = 1/2$) is not part of any claim. The methodology record (token economy: $\approx$700k tokens, 1009 requests, 99.2% context-cache retransmission) is reported as an efficiency case study, with the single data point that after a sleep interval the intuition compacts — recorded, not concluded.

First-completion times (Riemann direction). The visualization was delivered 2026-08-06 17:55 (request 17:10, UTC+8), with the complex-base extension completed 18:09 and the persisted data 18:46. The first Lean no-`sorry` of the whole trail is C001 (semiconjugacy), completed 2026-08-07 11:31 (first Lean commit 10:53). The Riemann-direction chain C011–C025 was completed 2026-08-12 00:00–02:54; the technical record and the full research paper were dated and published 2026-08-12 (technical record DOI 10.5281/zenodo.21896990, full paper 10.5281/zenodo.21897167, and the earlier DOI 10.5281/zenodo.21896345 added to the main paper at the user's instruction 09:43–09:44, after the user corrected the model's referent choice at 09:44: "no, the Riemann formalization's main paper"). The basepoint-drift result inside the Riemann direction is the first-completed instance the user cites for the "first completion counts" rule. The user's instruction at 2026-08-13 03:05 — "revise the Riemann paper wording: every claim about 'proving' is what DeepSeek insisted on" — corrected the record's attribution: the unproved statements are the API model's insistence, not the user's claim.

4 The homework-exercise series (2026-08-13 → 08-15)

Parallel to the experiments, the trail produced a series of "homework exercises" for the framework's foundation paper: re-observations of classical problems through the Pat lens, each with its own Lean formalization anchored only to framework theorems. The series ran 2026-08-13 14:19 through 2026-08-15 (25 reports: exercises I–XXII + Pat0 overview + summary).

Exercise I (P vs NP) started the series at 2026-08-13 14:19 (session "PAT-view P vs NP paper and Lean verification"). The user's corrections steered it through three iterations in 27 minutes:

- 14:24 — positioning correction: "not the computation chapter of the psyche paper, it is a homework exercise for the foundation chapter"; the exercise expands claims R152/R153 (P vs NP in the Pat frame);
- 14:31 — scope correction: "do not over-enumerate, meaningless; under this framework we trade speed for fuzziness and recover precision with structure"; the initial 9-theorem enumeration was deleted, leaving the 4 mechanism theorems that carry the single thesis: NP's Pat semantics = "use fuzziness for speed (existence via phase-locking extrapolation R150), use structure to recover precision (verification table: witness $\iff$ table entry)"; the full build stayed green;
- 14:45–14:46 — the user asked for a detailed cross-reference document, and that all P vs NP material live in the exercise's own subfolder (a self-contained package pattern that all later exercises follow);
- 14:48 — citation instruction: "cite as many living mathematicians, philosophers, linguists and scientists as possible", with the discipline that every citation is verified via web search (no citations from memory).

Exercises II–IX (each: paper ZH/EN + Lean + claim + README) were written during 2026-08-13 in parallel sessions; at 18:26 the user asked to publish all exercise papers and Lean files. Ten DOIs were published (2026-08-13 18:26–18:37, 43 files, all HTTP 201). At 19:01 the user extended the requirement: "up to exercise 14, all must have double-blind publication structure"; exercises X–XIV had just been created by the parallel sessions, and the double-blind packages (15 at that point) were built and published within minutes.

The exercise list (topic / anchor / claim). Each exercise is self-contained (paper ZH+EN, Lean snapshot, claim YAML, README) and anchors only to framework theorems:

- **I: P vs NP in the Pat view (R157, 08-13 14:19–14:46)** — expands R152/R153: NP's Pat semantics = "use fuzziness for speed (existence via phase-locking extrapolation R150), use structure to recover precision (verification table: witness $\iff$ table entry)"; 4 mechanism theorems, 0 sorry; the cross-reference document added 14:45–14:46.
- **II: the 1 of each number domain (R158)** — $\sqrt{2}/2$, complex, arbitrary n-ary, transcendental numbers: each domain's "1" as its basepoint; 8 theorems, 0 sorry (PatNumberOnes.lean, anchored to R154's $\sqrt{2}/2$ projection).
- **III: Riemann conjecture and twin primes revisited (R159)**.
- **IV: trigger-stripping experiment (self-reference collapse initiation)** — the mechanism by which unpaired self-reference collapses.
- **V: four interlock as the minimal self-consistent interlock (R160)**.
- **VI: interlock growth step and projection detachment (R161/R162/R163)**.
- **VII: five forces domain-by-domain Pat observation (R164)**.
- **VIII: Pat-native group representation theory (R165)**.
- **IX: physical space interlock structure (R166)**.
- **X–XIX: the seven millennium/classical problems re-observed** — Navier–Stokes (R167), Hodge (R168), BSD (R169), Goldbach (R170), parity and primes (R171), twin-prime conjecture (R172), twin-prime gaps as an oscillation-period axis (R173), Collatz (R176), ABC (R179), odd perfect numbers (R180).
- **XVIII: the real axis in the Pat mapping architecture (R178)** — the real axis as a projection of the complex rotation algebra.
- **XXI: continuum = closure of the Pat grid** and **XXII: composite numbers = multiple projection superpositions of primes** — the two summary observations connecting the series back to the continuum construction (R150/R151/R163).

Every exercise carries the structure "exercise / topic / Lean / DOI" (per the user's 08-14 12:47 re-qualification) and the declaration that it represents no mathematical conclusion, only an observation report based on Lean formalization; every Lean file compiles with 0 `sorry`.

Exercises XV–XXII continued on 2026-08-14 (R167–R180: Navier– Stokes, Hodge, BSD, Goldbach, parity and primes, twin primes, Collatz, the real-axis mapping architecture, ABC, odd perfect numbers). On 2026-08-14 12:47 the user re-qualified the whole series: every exercise must carry the structure "exercise / topic / Lean / DOI" and declare that it *represents no mathematical conclusion, only an observation report based on Lean formalization.* The series converged on 2026-08-15 with 25 reports, all with Lean 0-`sorry` formalizations.

5 The user's theoretical findings: the token-system theory (2026-07-29 → 08-14)

Before and during the E-series, the user built a theoretical layer about what training intuition is and how representations should be organized. The E-series is its empirical testing ground; the framework formalizes its surviving parts. This section records the findings themselves (verbatim, with timestamps), each with its later landing point in the experiments or the formalization.

1. **Induction and deduction are construction and intuition** (2026-07-29, thinking document): "I understand that feeling and expression are like induction and deduction — a pair of inverse structures; induction relies on construction, deduction relies on intuition. This means that even within the same expression system, receiving and expressing, feeling and executing, are very likely *not to use the same token* even in the same modality." Landing: the symbol/concept layer separation of the token system and E2 (symbol mapping direction).

2. **The problem is also an element** (2026-07-29 22:46): "the problem/operation and the data are isomorphic — both are elements, both are logic-token vectors." Landing: the four-question-type system (judgment / choice / fill-in / computation), where the question itself is an element sequence; later the whole sample-synthesis architecture.

3. **Formal reasoning is the position of tokens, not tokens** (2026-07-29 18:38): "formal reasoning is essentially not tokens, but *positions* of tokens; induction is the discovery of positions, deduction and abduction are the use of positions; and a position can be a coordinate in a matrix, not necessarily a graph relation" (the model had wrongly narrowed positions to graph relations; the user corrected: "who said position must be a graph relation? Can't it be a position in some matrix?"). Landing: the position-value representation study (E6) and the chain-length representation.

4. **The knowledge starfield is a high-dimensional Hilbert space, not a tree** (2026-07-30 correction): "my understanding of logical abstraction factorization was completely wrong; the real knowledge starfield is a high-dimensional Hilbert space, not a tree"; and "=/≡ is a set of multiple logic tokens in the space (the intension and extension of a concept); independent points are the properties (logic tokens) that compose things". Landing: the logic-token design discipline — a concept is a token set, an element is a composition, and a "collision" (two concepts sharing a token pattern) is an incomplete axis set, not a bug.

5. **Token cancellation** (2026-07-30): "only tokens that all input elements share *with the same value* are cancelled" (the user rejected the model's "cancel 0" shortcut: "why cancel 0? My requirement was: if all elements at the input layer share the same token, cancel that token; I never said cancel

0"). Cancellation is set intersection — pure compilation, no dimensional degeneration in high dimensions. Landing: the cancellation/collapse mechanics of the v2–v5 training pipelines.

6. **The sample-size information bound (Mastermind steps)** (2026-07-29 22:20–22:24): "data synthesis should guarantee that every logic token pairing is trained enough — like the four-color vs eight-color guessing game: you cannot limit both to the same number of steps"; "the minimum guessing-game steps: given eight colors and four positions, each guess reports how many colors and how many positions are correct" — the sample count has an information-theoretic floor. Landing: information-gain sampling (the halving experiment, 22:35; the first generalization 22:39:12) and later the E-series sample budgets.

7. **Weight compression** (2026-07-30 08:17): "I understand this is essentially training more fully with fewer logic tokens — what we are doing is equivalent to compressing weights" (registering the logic gates AND/OR/NOT/XOR as logic tokens: the gates share the boolean polarity axes, so one token set covers eight operators). Landing: the 47% $\rightarrow$ 79% $\leq$-generalization result when logic-gate co-training completed the parallel counterparts (07-30 08:45); the "compression via shared axes" principle behind the token vocabulary.

8. **Parallel logic tokens, not parallel elements** (2026-07-30 05:30): "once all logic tokens of an element are sufficiently trained through parallel logic tokens, the element generalizes even if never trained". Landing: the 55% vs 90% two-pole (leaving $\leq$ with a broken counterpart axis vs complete counterparts); the E-series two-pole designs.

9. **The convergence phenomenon as an overfitting question** (2026-08-14): the observed token-training convergence (loss 6.59 $\rightarrow$ 1.53) must be examined for whether it is overfitting; the author's exploration starts from "what changes under pure-intuition overfitting". Landing: the overfitting discussion in the observation report and the E-series recovery judgments.

10. **Directionality as a primitive notion** (2026-08-01): direction must have a precise mathematical/logical definition (directed/undirected), because directions determine which pairs can be declared (R136: directions must be declared in pairs $\{d, -d\}$, twice declared = asymmetry). Landing: the paired-declaration law and E18 (paired 5/9 vs one-way 1/9).

11. **Symmetry as dimensionality reduction, the restoration point = midpoint** (2026-08-15 06:00): "symmetry is dimensionality reduction"; the restoration point (midpoint) of a symmetric pair is a generation rule. Landing: R163 (endogenous continuum construction) and E14 (symmetry center not at 0; center drift extrapolates).

12. **Orthogonality is itself a symmetry** (2026-08-15 09:14) and **two symmetry groups localize a direction** (09:19, I7v): "Pat0's declaration

must be two groups of symmetries; any thought experiment based on a single dimension is wrong”. Landing: the 09:16–09:19 verification (double declaration 5/5 stable; direction localization 16/16 $\times$ 3 seeds) and the logical-completeness project (Section 8.1).

13. **The logical completeness theorem** (R031, C1–C7): the decoupling operator D (R1 exclusion + R2 cancellation + R3 layering) $\Longrightarrow$ completeness at any order/element/subject-object: perfect OOD. Landing: Section 8.1 (verification project 2026-08-15 09:14–09:47).

14. **Mechanics symmetries come in interlocked orthogonal pairs** (2026-08-16, R233, 29 theorems): each degree of freedom is a four-phase interlock pair; the symplectic structure $J^2 = -I$; the full embedding is 12 pairs $= \mathbb{C}^{12}$; the gravitational direction is asymmetric (structure layer has no privilege, action layer is one-way). Landing: the R233 formalization (29 theorems, 0 `sorry`) and the E18 mechanics- token experiment (paired declaration 5/9 vs one-way 1/9).

15. **The five-layer token system B/C/S/G/P** (2026-07-29 $\rightarrow$ 08-16): basepostulate (B) / concept (C) / symbol (S) / grammar (G) / presentation (P); the layers have distinct roles — B/C are the meaning layer (what things are), S/G are the expression layer (how things are written and arranged), P is the concrete notation (symbols with precedence and associativity). A token is one thing; the JSON files are projections of it. Landing: the whole v5 tokenizer and the three mapping- judgment theorems below.

16. **The three mapping-judgment theorems** (2026-08-16 16:17–16:23, the user’s correction re-located them): how to *judge* the multi-to-multi mappings between layers — (i) **symbol-norm theorem**: the judgment of the multi-to-multi mapping from symbols to logic and intuition (every target must be resolvable; ambiguity converges by context; unresolvable = non-norm); (ii) **presentation-legality theorem**: the judgment of the mapping from logic and intuition to presentation (every presentation target must be resolvable; numeric coincidence without declaration is illegal); (iii) **intuition- precision theorem**: the judgment of the mapping from intuition to logic and presentation (the intuition-token interface must be definitionally resolvable, must not impersonate a logic concept; precision is judged by two-pole comparison; polarity-confusion events are the canonical failure). A fourth theorem (symbol $\rightarrow$ presentation) completes the five-layer coverage. Landing: `MappingJudgmentTheorems.lean` (5 theorems, 0 `sorry`, 2997 jobs) with the experiment-coverage matrix (E6/E4/E5/E12/E13/E15/E14 + R065) — Section 8.2.

17. **Symbols are not values; numerals are intuition** (2026-08-15 06:54): “numeral is itself intuition; the true numeral is the successor chain from the basepoint; two successors from the basepoint are addition”. Landing:

E6 (structure-bearing representation) and the whole chain-length representation of the E-series.

18. **The head paradigm: same-kind/intension-extension/same-value** (2026-07-31): the head is determined by whether tokens are same-kind, and within same-kind by the same-value relation of their intension/ extension — not by hand-crafted heads. Landing: the head-router design of the v2 trainer and the token-level judgment discipline.

19. **Number-symbols need both reference and definition** (2026-08-01): the numeral symbol's signifier vs referent (the symbol [3] vs the concept 3 differ in token composition); every derived token needs a definition proposition, not only a derivation. Landing: the definition-layer discipline (E4/E5: definitions are invisible to training) and the token-system maintenance rules.

The theoretical layer and the empirical layer are kept separate in this record: the findings above are the user's claims (each with its verbatim timestamp), the E-series results are the witnesses, and the Lean theorems are the mechanical verification. No finding is promoted to a theorem without a Lean proof; no E-result is promoted to a law without its two-pole evidence.

6 The empirical trail: E1–E18, experiment by experiment

The series is design-driven: each experiment answers the question raised by the previous one. We document each experiment individually — design reason, procedure, result, and significance — so that the trail can be audited step by step. Configuration throughout: dim 64, layers 2, epochs 60 (unless noted), lr 10^{-3}, seeds $\{0, 1, 2\}$, pure next-token transformer, token-level judgment (full-sequence reconstruction; position-level accuracy is not trusted).

6.1 E1: extraction-channel ablation (ablate_channels)

(first success: 2026-08-15, experiment day)

Design reason. Before any law can be claimed from an OOD result, the judgment itself must be strict: "training intuition" is an unobservable construct, and every claim about it needs a decision procedure. The trail distinguishes three extraction channels through which a training result can arise — the *sample* channel (what the training samples carry), the *construction* channel (what the representation's construction rules carry), and the *arrow* channel (what the predefined A-layer arrows carry). The question is which channel is *decisive*: remove it, and the OOD result must change. This is a completeness-style elimination: a channel is called decisive only if its removal is witnessed by a two-pole change (1-OOD $\rightarrow$ 0-OOD or the reverse), never by a single observation.

Procedure. Domain: $\mathbb{Z}/7$ (mod 7), where the multiplicative inverse exists for every non-zero element — a non-degenerate point probe that the additive domain cannot provide. Baseline: all three channels present. Then each channel was removed one at a time, every combination over multiple seeds, training accuracy and OOD judged per seed.

Result. Only the sample channel is decisive: removing it changes the outcome; removing construction or arrows alone does not. The two-pole evidence: with samples present, OOD succeeds; with samples removed, it collapses — regardless of construction or arrow status.

Token-level judgment. All results reported as full-sequence reconstruction (sample-pair/total + position-level collapse localization); aggregate accuracy alone is not trusted.

Significance and declaration boundary. Structure still carries information (ontology: the representation's construction rules and arrows are what the samples are *about*), but extraction must pass through samples (epistemology: the training process reads samples, not structures). The claim is about *which channel decides the training outcome*, not about structure being useless — the two statements are separated so that a reader cannot conflate them.

6.2 E2: symbol mapping direction (sym_map_exp)

(first success: 2026-08-15, experiment day)

Design reason. The token system separates a symbol layer (how things are written) from a concept layer (what things are). The question: does the direction of the mapping matter? A bare symbol → concept mapping crosses layers; a concept → concept mapping stays within one layer. If training intuition is layer-sensitive, the two directions must behave differently; if not, the layer distinction is presentation-only.

Procedure. Two training setups, same vocabulary, same seed set: (a) symbol tokens mapped directly to concepts (cross-layer), (b) concepts mapped to concepts (same-layer). Training accuracy and OOD compared across seeds.

Result. Symbols must map to *structured* concepts; a bare symbol mapped directly (without a structured target) is unstable — the OOD collapses, while the same symbol mapped through a structured concept extrapolates.

Token-level judgment. The collapse is located at the symbol-to- concept junction (the first token after the symbol); the input head passes, the failure is the mapping itself.

Significance and declaration boundary. This fixes the symbol discipline used throughout the trail: a symbol is not a value, it points at a structured concept; ambiguity is resolved by context, never by the symbol alone. The claim is about mapping *direction and target structure*, not about the symbol layer being eliminable — presentation remains necessary for expression.

6.3 E3: pure-structure nesting foundation (succ_nest_exp)

(first success: 2026-08-15 06:48, OOD 36/36; preceded by single-operation failures 06:43 OOD 0/15, corrected to mixed training)

Design reason. Before any law about anchors, notation, or 0, the trail must answer the foundation question: is isomorphic nesting — a chain whose length is countable and whose parts are shared — learnable at all? If nesting is not learnable, nothing that follows can stand. The logic is layered: (i) numerals as basepoint successor chains, (ii) addition and multiplication as chain-length operations, (iii) 0 as the empty chain.

Procedure. 4 tokens (zero/succ/addition/multiplication), succ-nested representation, addition and multiplication with 0 fully enumerated, 5 seeds. Configuration: dim 64, layers 2, epochs 60, lr 10^{-3}.

Result. Convergence 1.000 × 5 seeds; OOD 36/36 (new-range chain lengths) and 0-combination OOD 60/60 × 4 seeds. The isomorphic- nesting foundation is stable.

Token-level judgment. Full-sequence reconstruction passes for all 36/36 and 60/60 samples; the per-position trace shows no collapse at any chain position (the chain is read left to right and reconstructed position-for-position).

User correction during the run (06:15–06:48). The first attempt trained a single operation with small samples and failed (OOD 0/15): "a single operation with small samples cannot be learned anyway" (the earlier perfect generalization used 4816 mixed samples). The fix — mixed training with the target screened — is itself a recorded correction of the API execution: the model had to be steered from single-operation isolation to mixed-representation training before the 36/36 result appeared.

Significance and declaration boundary. Chain-length structure is learnable and extrapolates within its representation. The claim is scoped to the succ-nested representation; it says nothing yet about other representations (that is E6) and nothing about anchors (that is E11).

6.4 E4: four definitions of 0 (zero_exp)

(first success: 2026-08-15 06:54, OOD 60/60)

Design reason. 0 can be defined in at least four ways: axiomatic (a constant), structural (the empty chain), intuitive (the basepoint itself), or relative (the basepoint of the current frame). The question: does the *definition* of 0 affect training? If the definition layer is invisible to training, the four definitions must train equivalently; if visible, they must diverge. This separates the definition layer from the representation layer — a distinction the whole framework depends on.

Procedure. Four definitions of 0 × multiple seeds, identical sample structure otherwise (succ-nested numerals, full 0-combination coverage), dim 64, layers 2, epochs 60.

Result. All four definitions train equivalently: 1.000 training accuracy, OOD 60/60 across seeds. No divergence in any seed.

Token-level judgment. The 0-combination OOD set ($0{+}n$, $n{+}0$, $0{\times}n$, $n{\times}0$, $0{\times}0$) passes full-sequence reconstruction position-for-position; no collapse at the 0 token.

Significance and declaration boundary. Training intuition is sensitive only to the representation form, not to the definition layer. The claim is scoped: definitions that *produce the same representation* are indistinguishable to training; it does not claim all definitions are equivalent in general — only that training cannot see the difference. This is the definition-layer invisibility law (law 5).

6.5 E5: 0-definition × succ-design matrix (zero_succ_exp)

(first success: 2026-08-15, 1.000 × 24 runs)

Design reason. E4 varied the definition of 0 alone. The design matrix completes the picture: 0's four definitions × succ's two designs (explicit concept token vs atomic counting) — 8 combinations. If both axes are invisible to training, all 8 must train equivalently; any single divergence localizes exactly which layer is visible.

Procedure. 8 groups × 3 seeds = 24 training runs, same sample structure (succ-nested, full enumeration), dim 64, layers 2, epochs 60.

Result. All 24 runs reach training accuracy 1.000; no group diverges. Succ's explicit/atomic design is as invisible as 0's definition.

Token-level judgment. Verified on full-sequence reconstruction across all 24 runs (per-run sample pairs reported); no position-level collapse in any group.

Significance and declaration boundary. The definition layer is invisible on both axes of the design matrix. Scoped claim: invisibility holds for the 8 tested combinations under succ-nested representation; representation changes (E6) are a different axis and are not covered here.

6.6 E6: structure-bearing vs bare-symbol representation (zero_symbol_exp)

(first success: 2026-08-15 08:36–08:37; 0-pole: third-representation OOD 0/32)

Design reason. E4/E5 showed the definition layer is invisible. The representation layer is the next candidate: is a position-value notation (implicit place value, nominal position tokens) equivalent to a true numeral (fully expanded successor chain)? The intuition to test: *structure must be carried by the representation, not assumed.* A bare-symbol representation assumes the structure implicitly; a true numeral expands it.

Procedure. Four representations A/B/C/D trained with identical sample content, then OOD-tested: A = application-chain (true numeral, the E3 representation), C = chain with value tokens; B/D = position-value / bare-symbol variants. 30 OOD probes per representation per seed.

Result. A and C: OOD 30/30 (stable). B and D: OOD 0/30 (collapse). The two-pole is sharp: position-value needs structural support; the true numeral is a basepoint successor chain and extrapolates.

Token-level judgment. The B/D collapse is located at the position layer (position 4–14 fails; input head and read-out pair pass) — a mechanism fingerprint that distinguishes format-layer failure from operation failure, invisible to aggregate accuracy.

User correction (08:36). The user's stripping discipline applied here: "before a complete ablation comparison, every OOD result must be assumed to be an intuition path; differences exist only through comparison" — the third-representation 0/32 was only reportable after the same-representation baseline 32/32 existed (both poles observed).

Significance and declaration boundary. Structure-bearing representation is necessary for extrapolation; bare symbols collapse. The claim is about *representation form*, not about the mathematical content: both representations denote the same numbers, only the learnability differs.

6.7 E7: 0-combination holdout (zero_hole_exp)

(first success: 2026-08-15 07:22, 40/40 + far-OOD 24/24)

Design reason. A genuine OOD judgment requires that the test set contain no sample analogous to the training set. Earlier series trained 0-combinations fully (E3–E5); this experiment holds out part of the 0-combinations (some $0\times n$ and $n\times 0$ pairs) and asks whether the 0-absorption law extrapolates to the held-out pairs.

Procedure. Partial 0-combination training (5 of the 0-samples present), held-out 0-combinations as OOD (40 probes), plus a far-OOD set (24 probes, $0\times 5 \ldots 0\times 9$ on chains never seen).

Result. Held-out 40/40; far-OOD 24/24. The 0-absorption law extrapolates from 5 0-samples to $0\times 5 \ldots 0\times 9$.

Token-level judgment. Full-sequence reconstruction passes for all 64 probes; the 0 token is reconstructed correctly at every position it appears (no position-level collapse).

Significance and declaration boundary. 0 is not a special exception in the representation; it is chain-length 0, and its rules extrapolate exactly like any other chain rule. The claim is scoped to the succ-nested representation; the "0 is special" hypothesis (needing special training) is rejected by the evidence, not by assumption.

6.8 E8: 0-multiplication vs nonzero multiplication (zero_vs_nz_exp)

(first success: 2026-08-15 07:25, 0-multiplication zero-shot 45/45)

Design reason. E7 showed 0-absorption extrapolates when some 0-samples are present. The sharper question: is 0 special at all? Train *only* nonzero multiplication (0-multiplication zero-shot), then test 0-multiplication. If 0 is chain-length 0, the multiplication rule must extrapolate to it without any 0-sample; if 0 needs its own training, it collapses.

Procedure. Nonzero-multiplication training only; OOD: all 0-multiplication pairs ($0\times n$, $n\times 0$, 0×0), 45 probes, multiple seeds.

Result. 0-multiplication zero-shot OOD 45/45. The chain-length multiplication rule naturally extrapolates; 0 is chain-length 0, not a special case.

Token-level judgment. All 45 probes pass full-sequence reconstruction; the 0 token at both operand positions reconstructs correctly (no position-level failure).

User instruction (06:54). This experiment exists because of the user's correction: "2. Then 0-multiplication must be compared against nonzero multiplication" — the model had to be steered from training 0-combinations inside the sample set to leaving them out entirely.

Significance and declaration boundary. The result is a law-level claim (0 = chain-length 0) with a zero-shot witness. Scoped: the claim holds for the succ-nested representation and the multiplication rule as encoded; it does not generalize to arbitrary operators by itself (E13 tests operation distinction separately).

6.9 E9: succ shift invariance (succ_shift_exp)

(first success: 2026-08-15 07:33, succ(1..7) 14/14)

Design reason. The chain-length reading of numerals assumes that succ(0)=1 and succ(1)=2 are *the same* succ applied to different chain lengths. If that is true, training succ(0)=1 alone (2 samples) must suffice for succ(1..7); if succ were position-dependent, it must fail. This is the shift-invariance question: is the successor rule translation- invariant in the representation?

Procedure. Train only succ(0)=1 (2 samples) and the 0-identity samples; OOD: succ(1..7) (14 probes), with a value-zero error-sample control group. Multiple seeds.

Result. succ(1..7) OOD 14/14. Shift invariance holds: 0→1 and 1→2 are the same succ. The error-sample control shows no difference (value-zero mistakes do not affect the shift result).

Token-level judgment. 14/14 full-sequence reconstruction; each succ chain position reconstructs correctly (the succ token at every position, the numeral at the end).

Significance and declaration boundary. The mechanism of chain- length extrapolation is shift invariance: the successor rule is translation-invariant, which is exactly what makes a chain a chain. Scoped: invariance is witnessed for one succ axis; multi-axis behavior is E10.

6.10 E10: multi-axis succ independence (axis_succ_exp)

(first success: 2026-08-15 07:33, same-axis 12/12)

Design reason. E9 showed one succ axis is shift-invariant. Does the invariance transfer across axes? Two succ axes (succ_a on axis a, succ_b on axis b) may share the rule or be independent. The question is answered by training one axis and OOD-testing the other: if axes share the succ rule, cross-axis OOD passes; if each axis has its own succ, it collapses.

Procedure. Dual-axis and single-axis training; single-axis ablation (train axis a only, OOD axis b). 12 probes per direction, multiple seeds.

Result. Same-axis 12/12; cross-axis 0/12. Each axis's succ is independent; a single axis can be ablated away without the other recovering it.

Token-level judgment. The cross-axis collapse is located at the first succ_b token (the axis-identity position); same-axis passes position-for-position.

Significance and declaration boundary. Succ is axis-relative: the rule is per-axis, not global. This is the first precise statement of "direction is declared per axis" — the empirical seed of the framework's paired-direction-declaration law (R136). Scoped: independence is witnessed for two axes; it does not claim a bound on the number of axes.

6.11 E11: basepoint drift (anchor binding) (basepoint_drift_exp)

(first success: 2026-08-15 07:43, same-anchor 32/32, drift 0/32; explicit bridging also 0/32)

Design reason. E9/E10 showed succ is per-axis. The deeper question: is the *operation* itself bound to its basepoint (anchor)? If operations are anchor-bound, moving the anchor must break the OOD (drift 0-pole), and adding an explicit bridge must fail to recover it; if operations are anchor-free, both must pass. This is the reference-frame question — the empirical root of the basepoint-relative operator theory.

Procedure. Anchor-0 and dual-anchor training; OOD three ways: same-anchor (baseline), drift (same operation, different anchor), explicit bridge (drift + declared equivalence between anchors). 32 probes per condition, multiple seeds.

Result. Same-anchor 32/32; drift 0/32; explicit bridging 0/32. Operations are anchor-bound: every reference frame needs its own empirical evidence; a declared bridge does not transfer the operation.

Token-level judgment. The drift collapse is located at the tail ("reads but cannot compute": input head passes 32/32, the computation positions collapse, 0.92 position accuracy at the tail) — a distinct mechanism fingerprint from E6's format-layer collapse.

User intuition (07:13–07:43). The user's direction steered the whole design: "the underlying choice is fine, the key is basepoint drift — what must be compared is basepoint drift; value_zero should be the basepoint's successor, right?" The experiment compares exactly that: drift vs same-anchor, with value-zero as the basepoint successor.

Significance and declaration boundary. This is the anchor-binding law (law 1): operations bind to their basepoint; each reference frame requires independent empirical evidence. The claim is about *training transfer under anchor change*, not about mathematical invariance — the mathematics of translation is trivially invariant, the training is not. This asymmetry (math invariant, training bound) is the phenomenon the framework formalizes.

6.12 E12: succ structure matrix, two-pole (succ-matrix experiment)

(first success: 2026-08-15 08:44 bridging 32/32; reproduction 2026-08-16 14:54–15:04 per-seed identical)

Design reason. E11 showed anchor binding. The same-basepoint question remains: two succs at the same basepoint (succ_a, succ_b) — are they the same operation in different notations, or different operations? The answer decides whether notation is translatable. The two-pole design is deliberate: *both* poles must be observed before any claim (two-pole discipline I7ag) — the zero-shot notation change (0-pole) and the equivalence-declaration bridging (1-pole).

Procedure. Five groups $\times$ 3 seeds = 15 runs (dim 64, layers 2, epochs 60):

1a baseline: train succ_a (250 samples), OOD succ_a;

1b zero-shot: train succ_a, OOD $\mathbf{succ}_b$ (same basepoint, notation change, no bridge);

2 bridging: succ_a + equivalence/anchor-error samples (270), OOD $\mathbf{succ}_b$;

2′ control: same 270 samples, OOD succ_a;

4 symmetry: $\mathrm{succ}_a/\mathrm{succ}_b$ mixed (252), OOD mixed.

Result. All 45 runs (5 groups $\times$ 3 seeds $\times$ 3 independent runs) reach training accuracy 1.000. OOD: 1a 32/32; **1b 0/32**; **2 32/32**; 2′ 32/32; 4 32/32. The two-pole (0/32 vs 32/32) never deviates across seeds or runs.

Token-level judgment. The 1b collapse is located at the notation-branching position (the first succ_b token); the input head passes (32/32 input reconstruction), the failure is the notation fork. Aggregate accuracy hides this; token-level reporting exposes the mechanism fingerprint.

User discipline applied. Two-pole pre-declaration (I7ag): a comparison may only be claimed when both poles are observed. 1b is the 0-pole, 2 is the 1-pole; the claim “equivalence declarations bridge notations” is licensed only by the pair.

Reproduction (2026-08-16 14:54). The user asked to reproduce the transformer experiments once more. E12 was re-run (15 runs), then run a third time inside the self-contained package; all three runs agree per-seed. The reproduction is part of the evidence chain, not a re-derivation.

Significance and declaration boundary. Training intuition is notation-bound statistics: 1.000 training accuracy does not transfer to an untrained notation; OOD 1.0 proves only that statistics extrapolate within that representation. Equivalence-declaration samples (pairwise declarations; anchoring errors irrelevant) bridge the two notations — declarations make the information point at the correct structure. This is the notation-translatability law (law 2) and the empirical seed of the phase-locking claim (R138).

6.13 E13: operation-distinction extrapolation (succ_disc_exp)

(first success: 2026-08-15 08:48, 56/56)

Design reason. Can the training intuition distinguish operations that share representation parts? The confusion pairs (2×2 vs 2+2, 0×2 vs 0+2) share every token except the operator. If operation distinction is carried by the operator token alone, the distinction extrapolates; if it depends on co-occurrence statistics, the OOD confusion pairs collapse.

Procedure. Standard training (mixed operations, full 0 coverage); OOD: confusion pairs (56 probes) on chain lengths/values inside the training range but never co-occurring with the tested operator in the training set.

Result. 56/56. Operation distinction extrapolates within the representation: the operator token carries the distinction, and the confusion pairs are resolved at every probe.

Token-level judgment. 56/56 full-sequence reconstruction; the operator position and the result position both reconstruct correctly on every confusion pair.

Significance and declaration boundary. This is the 1-pole of the distinction question. Its 0-pole is E16 (no fake samples: 14/28, 50%, truth bias) — the pair of experiments, not either one alone, establishes the claim "distinction requires the operator's negative evidence". Scoped: within-representation distinction extrapolates; cross- representation distinction is a different question (E15).

6.14 E14: symmetry center not at 0 (sym_center_exp)

(first success: 2026-08-15 06:00–06:16 midpoint OOD 1.000, epoch 13 (R163); formal re-run 2026-08-16, all OOD 100% × 3 seeds)

Design reason. The user's observation: "some symmetries have their axis not at 0 yet are natural numbers" — for a symmetric pair $\{a, b\}$, the center $c = (a + b)/2$ is a natural number; the symmetry axis (center) drifts from 0 to natural numbers. Does the training intuition follow? This connects to R163 (symmetry as dimensionality reduction: restoration point = midpoint) as a generation rule for the continuum construction.

Procedure. Midpoint operation (midpoint token, chain-length representation), three groups × 3 seeds: (1) full-range extrapolation (train $a, b \in [0, 8]$, OOD $[9, 11]$); (2) center drift (train $[0, 6]$, OOD centers ≥ 5, 72 probes); (3) symmetric-pair restoration (train centers ≤ 4, OOD centers 5..7).

Result. All OOD 100% (10/10, 72/72, 24/24 × 3 seeds; training accuracy 1.000). The symmetry axis (center) is *not* an anchor: the center is a computable result of chain-length arithmetic $((\text{len}(a) + \text{len}(b))/2)$; with the structure fully expanded, the center's drift from 0 to any natural number extrapolates.

Token-level judgment. Full-sequence reconstruction passes for all probes; the midpoint token and the center numeral reconstruct position-for-position in every group.

Contrast with E11 (both poles). Basepoint (chain bottom) = anchor (drift collapses 0/32, E11); midpoint (symmetry axis) = generation rule (drift stable). The two kinds of "axis" are essentially different — a distinction invisible to aggregate accuracy, exposed by the two-pole comparison.

Significance and declaration boundary. R163's continuum construction (dyadic midpoint generation) is empirically extrapolable in the transformer. The claim is scoped to chain-length representation; cross-representation transfer is tested separately (E15).

6.15 E15: strict experiment (strict_center_exp)

(first success: 2026-08-16, zero-shot 26/26; cross-representation 0/26)

Design reason. E14's honesty boundary: its group-2 OOD partially overlapped trained a, b values (combination extrapolation). The strict version: fully zero-shot center drift (chain lengths never trained), plus the cross-representation two-pole (I7ag: one 0-pole and one 1-pole before claiming).

Procedure. Four groups × 3 seeds:

1 fully zero-shot center drift (train $a, b \in [0, 4]$, OOD $[5, 9]$, chain lengths never trained);

2 cross-representation stripping (train chain-length, OOD position-value notation);

3 cross-representation bridging (chain-length + numeric-equivalence samples, OOD position-value);

4 symmetric pairs fully zero-shot (train centers ≤ 3, arms ≤ 2; OOD centers 4..6, arms 3..4).

Result.

Group	OOD	token-level	judgment
1 zero-shot center drift	26/26 ×3	793/793 (1.00)	chain-length arithmetic zero-shot
2 cross-repr. stripping	0/26 ×3	156/624 (0.25)	0-pole: format layer (pos 4–14)
3 cross-repr. bridging	15–25/26	unstable	new finding: numeric equivalence $\neq$ operatio
4 zero-shot symmetric pairs	12/12 ×3	—	center+arms fresh, extrapolated

Token-level judgment. Group 2's collapse is located at the format layer (positions 4–14 all collapse; input head and read-out pairs pass) — the same fingerprint as E6's B/D collapse, confirming the format-layer mechanism across experiments.

New finding (group 3). Numeric equivalence does not equal operation equivalence: heterogeneous-representation bridging requires operation-level equivalence samples. Same-isomorphism notations bridge fully (E12); heterogeneous representations need operation-level declarations (E6's isomorphism/heterogeneity boundary reappears at the bridging dimension).

Significance and declaration boundary. The two-pole holds (1 and 4 at the 1-pole, 2 at the 0-pole): "the symmetry axis extrapolates" is strictly scoped to the representation-internal intuition path — zero-shot extrapolation (chain-length arithmetic) succeeds, cross- representation collapses (format layer), and equivalence samples bridge only partially (operation-level equivalence required).

6.16 E16: distinction-task 0-pole, no fake samples

(first success: 2026-08-16, 14/28 = 50%)

Design reason. E13 established the 1-pole of operation distinction (56/56 under standard training). The coverage audit exposed a single-pole problem: standard training includes negative evidence (fake results), so 56/56 could be carried by the negative samples alone. The 0-pole: train *without* fake samples (only true results), then OOD the confusion pairs. If distinction requires negative evidence, the no-fake training must collapse to chance.

Procedure. Training: true samples only (no fake results). OOD: the same confusion pairs as E13 (28 probes).

Result. 14/28 = 50% — chance. Without conflicting information the model cannot judge distinction (truth bias: the model defaults to "true" when it cannot distinguish).

Token-level judgment. The collapse is symmetric across the confusion pairs: both directions (0×2 vs $0+2$ and their reverses) fail at the operator position; no position-level asymmetry.

Significance and declaration boundary. Together with E13, the two poles are complete: distinction requires the operator's negative evidence; standard training provides it, no-fake training does not. The claim is about *evidence composition*, not about model capacity — the same model, same vocabulary, same sample count, differs only in the presence of fake samples.

6.17 E17: two-pole pre-declaration (methodology upgrade)

(first success: 2026-08-16, both poles complete)

Design reason. E13/E14's single-pole problem was a process flaw: the two-pole discipline (I7ag) was established *after* those experiments were run, so their 0-poles had to be added retroactively (E16). E17 upgrades the methodology: the 0-pole and the 1-pole are *pre-declared at design time*, before any training runs.

Procedure. Three groups, each with a pre-declared 0-pole and 1-pole (six conditions), × 3 seeds:

a equivalence: chain-length vs bare-symbol representation of equivalence (1-pole: chain-length; 0-pole: bare symbol);

b distinction: true+fake vs no-fake training on the confusion pairs (1-pole: 28/28; 0-pole: 14/28);

c center: chain-length midpoint vs cross-representation midpoint (1-pole: 10/10; 0-pole: 0/26).

Result. Every group shows the full two-pole column: a 0-pole collapse and a 1-pole pass, both pre-declared. (a) 50/50 vs 2/50; (b) 28/28 vs 14/28; (c) 10/10 vs 0/26.

Token-level judgment. Each pole reported token-level (sample pairs + position-level collapse localization) per the reporting discipline; no aggregate-only claims.

Significance and declaration boundary. The methodology law: a comparison is claimed only when both poles were declared before the runs and both were observed after. The upgrade is itself part of the trail — the discipline's history (E13/E14 retroactive, E16/E17 pre-declared) is recorded, not erased.

6.18 E18: mechanics-token full set

(first success: 2026-08-16, paired 5/9 vs one-way 1/9)

Design reason. The framework claims directions must be declared in pairs (R136). The experiment tests this at the token level: 8 mechanics tokens (succ/add/sub/mul/div/mid/recip/inv) in the same chain-length representation (single structure, not mixed — the E17a representation lesson). Two training regimes: paired declaration (inverse pairs trained together) vs one-way (only the forward direction). OOD: new-range confusion pairs over all tokens.

Procedure. 1-pole: paired inverse training; 0-pole: one-way training. OOD identical (new-range confusion pairs, all-token pairs).

Result. Inverse-knowledge discrimination 5/9 (paired) vs 1/9 (one-way): paired declaration makes difference decidable, one-way collapses — R136's empirical witness. Forward extrapolation passes in both regimes (8/8, chain-length statistics). recip/inv fail in both regimes (4 probes): 2 samples are too few (EXP-SYM-INV replication) plus structural poverty of the natural-number domain.

Token-level judgment. The paired advantage is located at the inverse-pair positions (the discriminator needs both directions to separate the pair); forward positions pass in both regimes.

Significance and declaration boundary. Direction declarations are empirically effective in pairs; single declarations leave the discrimination undecidable. The claim is scoped to the 8 tested tokens and the chain-length representation; recip/inv's failure is a data-diversity statement (2 samples), not a token-design claim.

7 Five structural laws

The E-series converges on five laws, each with its two-pole evidence:

1. **Anchor binding (E11, first success 2026-08-15 07:43).** Operations bind to their basepoint; reference frames require independent evidence

(same-anchor 32/32, drift 0/32, bridging 0/32). Math is invariant, training is bound — the asymmetry the framework formalizes.

2. **Notation translatability (E12, first success 2026-08-15 08:44).** Same-basepoint notation change collapses zero-shot (0/32) and bridges with equivalence declarations (32/32); notation is translation only via declarations. (R138 phase locking.)

3. **Shift invariance (E9, first success 2026-08-15 07:33).** The successor rule is translation-invariant within an axis (succ(0) alone yields succ(1..7) 14/14); extrapolation is chain-length arithmetic.

4. **Structure-bearing representation (E6, first success 2026-08-15 08:36–08:37).** Position-value collapses (0/30), true numerals extrapolate (30/30); structure must be carried by the representation, not assumed (E7/E8: 0 = chain-length 0, zero-shot 45/45).

5. **Definition-layer invisibility (E4/E5, first success 2026-08-15 06:54).** The definition of 0 and the design of succ are invisible to training (24/24 runs equivalent); training reads representation form, not definitions.

Each law is stated with its polarity evidence and its scope; no law is claimed beyond its representation (chain-length, dim 64, layers 2).

8 From laws to formalized framework

The laws map to the Pat framework (basepoint-relative operator semantics):

Law	Framework claim	Lean file (0 sorry)
Anchor binding	R136 directions declared in pairs	
Notation translatability	R138 phase locking	MutualLocking.lean (R143)
Equivalence declaration	R143 interlock matrix det $\neq 0$	MutualLocking.lean
Basepoint = anchor	basepoint-relative operators	BasepointMove.lean (C026)
Continuum	R150/R151/R163	PatCountableInfinitPhaseUnification.lean ContinuumPatGrid.lean, ContinuumConst

The formalization: claims R136–R153 (the framework main chain) all PROVED with no `sorry`, completed 2026-08-13 13:31 (18 claims); the Toolkit branch extends to 150+ Lean files, claims R001–R232 (with 43 Ruler* experience-layer claims and TK1–TK5); full `lake build` passes (3631 jobs, mathlib v4.32.2, lean-toolchain pinned). Key theorems: $J^2 = -I$ (symplectic rotation, R149 four-phase interlock, S^3), the mechanics-symmetry refinement R233 (2026-08-16, 29 theorems, 0 `sorry`): the four layers of mechanics symmetry, each interlocked symmetry pair orthogonal, the symplectic structure ($J^2 = -I$; each degree of freedom = a four-phase interlock pair), the full symplectic embedding (12 pairs = $\mathbb{C}^{12}$), and the gravitational symmetry direction (structure layer has no privilege $\wedge$ action layer is one-way), mutual locking $\iff$ determinant $\theta(r + 1/r) \neq 0$ (R143), the countable-infinity/continuum phase-locking unification (R150), the continuum as Pat-grid closure (R151), and the endogenous continuum construction with zero Nat in the definition layer (R163). The ZeroRelative branch formalizes the basepoint-relative algebra (C001–C025: heap retract, semiconjugacy, displacement equivalence, endogenous generation) and the operator semantics (C026–C038: every operator has its own basepoint, ± 1 as projection extremes, symplectic projection, Q2_0 grid as sign encoding, state-matrix complex structure).

9 The logical completeness theorem and the three mapping-judgment theorems

9.1 The logical completeness theorem (2026-08-15 09:14–09:47)

At 09:14 the user asked to review the whole process against the framework's logical completeness theorem and to design a formal verification project: "how do you run a logically complete experiment? And I conjecture that orthogonality itself is a symmetry — can a single axis complete training? That needs to be compared." The theorem, as extracted from the claims ledger (R031, completeness C1–C7), states:

> *The decoupling operator D (R1 exclusion + R2 cancellation + R3 layering) $\Longrightarrow$ completeness at any order, any element, any subject-object: perfect OOD.*

The verification project (same day):

- 09:16 — orthogonality-declaration bridging succeeds same-anchor (32/32); cross-anchor orthogonality is seed-dependent (B/C: 32/32, 32/32, 0/32), in contrast to the same-anchor stability of E12;
- 09:17 — the *double* declaration (orthogonality + equivalence) restores stability: 5/5 across seeds. A single declaration is insufficient;
- 09:18—09:19 — the user drew the corollary (I7u): if orthogonality is a symmetry, there is no axis orthogonal to every axis; an "orthogonal axis" is a phase on an orthogonality symmetry axis. Then the stronger claim (I7v): *two groups of symmetries are necessary to localize a direction* — Cartan–Dieudonné: rotations are composites of two reflections, a single reflection only defines $\pm$; therefore Pat0's declarations must be two groups of symmetries, and any thought-experiment based on a single dimension is invalid. Direction localization with two symmetry groups: OOD 16/16 $\times$ 3 seeds (09:19);

- 09:22 — the user asked for the minimum number of symmetry groups when the declared directions are not two orthogonal axes (conjecture: 3 or 4, because an orthogonal phase implies an angle mapping);
- 09:30–09:36 — the completeness dimensions (order / element / subject-object) re-verified token-level per the new reporting discipline (algebraic axioms 12/12, involutions 12/12, truth/false separation 6/6);
- 09:19 — the methodology was written up as the short paper *Logical-Completeness Experimental Design Methodology*, later merged into the release package ("self-redemption series").

9.2 The three mapping-judgment theorems (2026-08-16 16:16–16:23)

On 2026-08-16 the judgment theorems for the token system were located as *multi-to-multi mapping judgments*. The user's correction at 16:17 fixed the two-theorem positioning: "the symbol-norm theorem is: how does one judge the multi-to-multi mapping from symbols to logic and intuition; the presentation-legality theorem is: how does one judge the multi-to-multi mapping from logic and intuition to presentation." The three theorems:

1. **Symbol-norm theorem** (symbols → logic/intuition): every mapping target must be resolvable — ambiguity converges by context (no context = all candidates; given context = one); unresolvable ambiguity = non-norm. Two symmetric directions (A: symbol → logic; B: symbol → intuition).
2. **Presentation-legality theorem** (logic/intuition → presentation): every presentation target must be resolvable (notation context: precedence/associativity, concrete notation in the P layer); numeric coincidence without declaration is illegal (equivalence bridging must be declared). Two symmetric directions.
3. **Intuition-precision theorem** (intuition → logic/ presentation): the intuition-token interface must be definitionally resolvable (does not impersonate a logic concept); intuition-path precision is judged by two-pole comparison; polarity-confusion events are the canonical failure.

A fourth theorem (symbol-presentation mapping) was added at 16:19–16:20, completing the five-layer mapping system (B/C/S/G/P). All four were formalized in `MappingJudgmentTheorems.lean` (5 theorems, 0 `sorry`, `lake build` 2997 jobs) with an experiment-coverage matrix (E6/E4/E5/E12/E13/E15/E14 + R065), committed 16:23.

10 Timestamped evidence chain

The trail is anchored by six independent evidence sources (all in the artifact):

1. Session records (db.sqlite message/part, per-message timestamps, UTC+8; 55,047 messages, 2026-07-15 onward) — the primary source; every timestamp in this paper is traceable to a session record.

2. Task execution log: 2303 turns (turn_usage; later extended to 2773).

3. Model-viewpoint phase records (daily) — the API model's own claims separated from the user's, per the distillation-pressure corrections.

4. git log commit timestamps (first Lean commit 2bc157e, 2026-08-07 10:53; first Lean no-`sorry` C001 at 11:31 the same day; Riemann-direction archive ec87658 2026-08-12 02:54; full-project track df65309 2026-08-13 17:48).

5. Experiment artifact file timestamps (E1–E13 results, 2026-08-15 07:22–09:00; result files in the self-contained package, cross-checked against session records).

6. Lean no-`sorry` timeline (R136–R153 completion 2026-08-13 13:31; 786 no-`sorry` records extracted and deduplicated on 2026-08-16).

7. The four-phase classification record (2026-08-16): all 4068 user messages from 2026-07-29 classified as positive / leaning-positive / neutral / leaning-negative / negative (13 / 101 / 2159 / 37 / 1758 after deduplication to 2158) — an independent audit of which statements were the user's affirmations and which were the model's.

Key anchors (each with its session record): 2026-07-29 22:39:12 (UTC+8) first *imperfect* generalization (halving experiment, 450 samples); 2026-07-29 22:52 strongest early generalization (train 5 relations leave 1: $\leq 81\% \to 90\%$); 2026-08-06 17:10–18:46 Riemann visualization (delivery 17:55); 2026-08-07 10:53/11:31 Lean first commit / first no-`sorry`; 2026-08-12 00:00–02:54 Riemann formalization session (C011–C025); 2026-08-12 framework core claims (R134–R153) and the Riemann publication (09:43–11:56); 2026-08-13 13:31 all 18 framework claims PROVED; 2026-08-13 14:19–18:37 exercise series start and first publication (10 DOIs); 2026-08-14 12:47 exercise re-qualification; 2026-08-15 04:01 first *perfect* generalization; 2026-08-15 07:22–09:00 E1–E13; 2026-08-15 09:14–09:47 logical-completeness verification project; 2026-08-16 14:54–15:04 E12 reproduction; 2026-08-16 16:16–16:23 three mapping-judgment theorems; 2026-08-16 self-contained package.

11 Human corrections steering the API execution

The trail is also a record of how the user's instructions corrected the API model's execution. We list the documented corrections (each with its session timestamp) because they are part of the evidence: they show which statements are the user's claims and which are the model's insistence.

- **2026-08-12 09:44** — referent correction: "no, the main paper of the Riemann formalization". The model had picked the wrong document (Unified Framework instead of the Riemann main paper); the user disambiguated; the DOI was added to the right paper.

- **2026-08-12 11:08** — attribution correction: the user's instruction that the Riemann paper must state explicitly: no claim of proving the Riemann hypothesis / all results are restatements of known facts / novelty = KNOWN. (Later reinforced at 2026-08-13 03:05: "revise the Riemann paper wording: every claim about 'proving' is what DeepSeek insisted on" — the unproved statements are the model's insistence, not the user's claim.)

- **2026-08-13 14:24** — positioning correction (exercise I): "not the computation chapter of the psyche paper, it is a homework exercise for the foundation chapter".

- **2026-08-13 14:31** — scope correction (exercise I): "do not over-enumerate, meaningless; under this framework we trade speed for fuzziness and recover precision with structure" — the 9-theorem enumeration was deleted, leaving 4 mechanism theorems.

- **2026-08-13 14:48** — citation instruction (exercise I): "cite as many living mathematicians, philosophers, linguists and scientists as possible", with every citation verified (no citations from memory).

- **2026-08-14 12:47** — series re-qualification: all exercises must carry "exercise / topic / Lean / DOI" and declare they represent no mathematical conclusion, only observation reports based on Lean formalization.

- **2026-08-15 09:14** — completeness-check instruction: review the whole process against the logical completeness theorem and design a formal verification project; conjecture that orthogonality is itself a symmetry (single-axis training must be compared). Generated the 09:16–09:19 double-declaration result and the two-symmetry-groups corollary (I7v).

- **2026-08-15** — two-pole discipline (I7ag): a comparison may only be claimed when *both* poles are observed (one 0-OOD and one 1-OOD). E13's single-pole design was flagged by this rule and re-done as E16/E17.

- **2026-08-15** — token-level reporting discipline: aggregate accuracy (0.99/1.0) is not sufficient; every claim must report full sample logs, token pair/total counts, and position-level accuracy (collapse localization). token_report.py was built under this rule.

- **2026-08-15** — intuition/structure stripping discipline: before a complete ablation comparison, every OOD result must be assumed to be an intuition-path result; structural stripping is mandatory.

- **2026-08-16 16:17** — theorem-positioning correction: "the symbol-norm theorem is the judgment of the multi-to-multi mapping from symbols to logic and intuition; the presentation-legality theorem is the judgment of the multi-to-multi mapping from logic and intuition to presentation". The three mapping-judgment theorems were re-located under this definition.
- **2026-08-16 06:17** — record-format correction: "you need to improve this: the time the motivation was proposed, the time the experiment was completed (first completion counts, e.g., the basepoint drift in the Riemann direction), and the time the paper was written". This rule generated the per-experiment first-success timestamps used throughout this paper.
- **2026-08-16** — timeline correction: the first generalization is 2026-07-29 (22:39:12 UTC+8); the model had misdated it; the user corrected the record.
- **2026-08-16 06:18 / 06:21** — attribution correction under distillation pressure: "you have already claimed it as your own, you know?" and "it is not just this: you have already published what we discussed". These corrections fixed the record's authorship boundary.
- **2026-08-16 14:54** — reproduction instruction: "reproduce our transformer experiments once more" — E12 re-run (15 runs, per-seed identical) + third in-package run.

12 The user's questions, corrections, and examples

The trail is driven by the user's questions. This section records them verbatim (with session timestamps), because they are the actual research hypotheses — each question determined the next experiment, and each correction fixed the record. The API model's role was execution and mechanical verification; the user's role was direction, judgment, and example.

12.1 Questions that steered the trail (verbatim, from the session log)

- **2026-07-29 18:46** — the first experiment: "we train an addition model, without providing the definition of addition, only the computation cases within 1000, and see whether 5-digit addition can be learned" (the pure next-token baseline: 2-digit 100%, 3-digit 0%).
- **2026-07-29 22:20** — the information bound: "the sample count should be calculated by the minimum guessing-game steps" (Mastermind mechanism; led to the information-gain sampling and the halving experiment).

- **2026-07-29 22:39** — the first generalization confirmation: "this means generalization happened, the necessary number of steps was not needed to get the necessary result. What is your halving logic, and which logic tokens generalized and which failed?" (the first per-token analysis, 22:41).
- **2026-07-30 00:00** — the research roadmap: "in math and logic, propositions are consistent (no self-reference, no contradiction); once we enter medicine, social science, law, there are many contradictory propositions that must live in one framework. This needs research — but first we solidify math and logic, by code practice."
- **2026-07-30 00:42** — the base-change conjecture: "I suspect that base-change equivalence naturally carries multi-digit out" (base change $\to$ addition generalization 81%, multiplication 12% — the latter requiring its own training, as the user then corrected: "of course multiplication must be trained").
- **2026-07-30 05:30** — the parallel-token principle: "note it is parallel *logic tokens*, not parallel elements — once all logic tokens of an element are sufficiently trained through parallel logic tokens, the element generalizes even if never trained" (verified: leaving $\leq$ with a broken counterpart axis 55% vs 90% with complete counterparts).
- **2026-08-12 00:14** — the honesty question: "wait, so along this line of thought can the Riemann hypothesis be proved?" (the model answered: no; all six theorems are OBSERVATION/KNOWN restatements, none touches the zeta function or zeros — the honest boundary is part of the record).
- **2026-08-12 00:16** — the true-complex-axis demand: "we cannot construct the prime axis, but we can construct the complex axis from lambda calculus, right? Note I want the *true* complex axis, not the fake one where each successor is the natural 1" (the user rejects axes whose successor is pre-assumed to be the natural numbers; the true axis is multiplicative iteration — rotation-scaling orbits).
- **2026-08-12 00:17** — the projection hypothesis: "for the construction of roots: suppose the complex axis is essentially the projection of the higher-dimensional structure -1 into human mathematical space; under this assumption, can it be constructed?" (formalized as the $a + bJ$, $J^2 = -1$ rotation algebra, 00:19–00:21).
- **2026-08-12 00:23** — the basepoint-drift statement: "since our constructed complex plane carries the naturals on its axis — do you know what this means? It means our lambda basepoint has moved onto the complex axis, i.e., the basepoint has drifted; its position is not the origin but some transform of i".

- **2026-08-12 00:25** — the fake-real-axis statement: "then that so-called real axis is very suspicious; I suspect it is not a real number axis at all, but another fake complex axis, right?".
- **2026-08-15 06:54** — the numeral intuition: "numeral is itself intuition; the true numeral is the successor chain from the basepoint ... and 0-multiplication must be compared against nonzero multiplication" (E8's zero-shot design).
- **2026-08-15 07:13** — the drift comparison: "the underlying choice is fine; the key is basepoint drift — what must be compared is basepoint drift. value_zero should be the basepoint's successor, right?" (E11's design).
- **2026-08-15 08:36** — the stripping discipline: "before a complete ablation comparison, every OOD result must be assumed to be intuition; differences exist only through comparison".
- **2026-08-15 09:14** — the orthogonality conjecture: "review the whole process against our logical completeness theorem, design a formal verification project on how to run logically complete experiments. Then a conjecture: I suspect orthogonality itself is a symmetry — can a single axis complete training? It must be compared."
- **2026-08-15 09:19** — the two-symmetry-groups corollary: "orthogonality as symmetry yields the conclusion: two groups of symmetries are necessary to localize a direction; therefore Pat0's declaration must be two groups of symmetries; any thought-experiment based on a single dimension is wrong" (verified: direction localization 16/16 $\times$ 3 seeds).
- **2026-08-15 09:22** — the minimum-groups question: "if the declared directions are not two orthogonal axes, how many groups of symmetries are needed at minimum? I suspect 3 or 4, because if an orthogonal symmetry has phase, then angle has a mapping".
- **2026-08-16 14:54** — the reproduction instruction: "reproduce our transformer experiments once more".

12.2 Examples citing the mathematicians' consensus

The user's examples often cite established mathematical consensus and then re-interpret it under the projection hypothesis — the pattern of the whole trail:

- **2026-08-07 (thinking document), translation vs inversion**: "mathematicians have told me that 1 and $1/a+1/b$ are surely symmetric, translation and inversion; I believe their construction. But under my hypothesis, $1/a + 1/b$ is the projection of inversion-1 after the projection lost structure. If my hypothesis holds, it can only mean one thing: the basepoint has drifted — the lambda-calculus origin has moved around the circle of radius 1 centered at 1". The consensus (the symmetric pair) is preserved;

the *mechanism* (why they are symmetric) is re-derived from projection loss.

- **2026-08-07 (thinking document), irrationals**: "I firmly believe irrationals are the remainder of some higher-dimensional structure after projection into our space lost part of the information-carrying structure. The reason is simple: every intuition-path construction moves along a direction away from the basepoint; only irrationals need to walk back. The unit ball in the higher dimension is complete 1 in one direction; projected into our world it leaves only $\sqrt{2}/2$ — this is likely the irrational's 1".

- **2026-08-09 01:27 (ChatGPT conversation), Church numerals**: "the lambda-calculus basepoint self-application iteration presupposes a concept distinct from the basepoint: the self-application operation, $\text{lambda}(\text{base}, \text{base})$. But in the concept I want to experiment with, there is no such operation: $\text{base}(\text{base}, \text{base})$" — the Church-numeral consensus is rejected as impure: it smuggles an implicit operation role, while the pure-relative design occupies operator/operand/basepoint with one primitive.

- **2026-08-15 09:19, Cartan–Dieudonné**: the user's two-groups corollary was checked against the classical theorem that rotations are composites of two reflections (a single reflection defines only $\pm$); the consensus confirms the corollary rather than the framework being derived from it — the direction-localization experiment ($16/16 \times 3$) is the witness.

12.3 Corrections during experiments (verbatim)

Beyond the record-format and attribution corrections of Section 9 (Human corrections), the experiment log contains execution corrections that directly changed designs:

- **2026-07-29 22:42** — output-format correction: "my requirement was to output the *element's logic tokens*, not the element! And this means our prefix question format needs more space" — the model had been outputting truth values; the four-question-type system followed (22:49).

- **2026-07-29 22:54** — architecture correction: "the data synthesizer, result translator, and result analyzer must all be pluggable; changing one position in the concept token takes effect globally" — the JSONL-driven CLI architecture.

- **2026-07-30 08:00** — "legacy's logic tokens are incomplete; the information about '=' was not trained/sampled — do you think this needs a logic token or training samples?" (both, answered by the experiment: 47% $\rightarrow$ 79% with completed counterparts).

- **2026-08-15 06:15** — "a single operation with small samples cannot be learned anyway" — the failed single-operation run (OOD 0/15) was corrected to mixed training before E3's 36/36.

- **2026-08-16 16:17** — the theorem-positioning correction for the three mapping-judgment theorems (Section 8.2).

13 Reproducibility and provenance

The artifact is a self-contained package (also published as a standalone repository):

- Training: E12 one-command reproduction (5 groups × 3 seeds, GPU auto, 5–10 minutes); E17 two-pole pre-declaration; E15 strict experiment; full E1–E13 (30 minutes). All results re-generate the package's report tables.
- Formalization: `lake build` (pinned lean-toolchain v4.32.2 + mathlib v4.32.2), 0 `sorry`.
- E12 reproduction fidelity: three independent runs (original ×2 + in-package) per-seed identical, no random drift.
- Provenance methodology: SHA-256-anchored release manifests, per-claim SHA256 publication, per-claim anonymous mirror (dual-repo: real + anonymized), and a publish pipeline (scan → SHA-256 digest → manifest → real repo → anonymize → blind repo).

14 Related work and conclusion

The empirical part relates to interpretability and OOD generalization studies of small transformers (known: OOD extrapolation depends on representation; cf. [**?**, **?**]); the formal part builds on heap/torsor algebra [**?**] and the previously submitted Riemann-direction formalization (C011–C025). The trail as a whole — observation → laws → framework → machine-checked theorems, with timestamped evidence — is the contribution.

Conclusion. We record a complete research trail: from the 0/1 OOD dichotomy of a low-epoch transformer (2026-07-29, 22:39:12; E12: 0/32 vs 32/32 across 45 runs) through eighteen individually documented experiments, five structural laws, and the Pat framework, to a Lean formalization (R136–R153 all PROVED, no `sorry`; 3631 jobs; R150/R151/R163 continuum chain). Three parallel threads are part of the same trail: the Riemann direction (2026-08-06 visualization → 2026-08-12 C011–C025 formalization and publication), the homework-exercise series (2026-08-13 → 08-15, 25 reports, all Lean 0-`sorry`), and the verification methodology itself — the logical-completeness project (2026-08-15 09:14–09:47: decoupling operator D = R1 exclusion + R2 cancellation + R3 layering, completeness at any order/element/subject-object, the two-symmetry-groups corollary) and the three mapping-judgment theorems (2026-08-16 16:16–16:23, `MappingJudgmentTheorems.lean` 0 `sorry`).

The trail is timestamped by seven independent evidence sources and reproducible from a self-contained package. Its writing follows the recording disciplines of the trail itself: every passage states its premises before its conclusion (no logical leaps: a claim is licensed only by its two-pole evidence); every symbol, number, and timestamp is traceable to a session record (no unanchored notation); every statement carries its provenance — the user's instruction, the model's insistence, or the mechanical verification — (no unverifiable assertion); and the user's intuitions are recorded separately from the API model's interpretations, with the corrections that separated them (no conflation of intuition path and logic path). Under provenance pressure, we argue, recording the *path* — not only the destination — is the defensible form of priority: each experiment, each claim, and each theorem carries its own timestamped anchor, and each anchor is either a session record, a commit, a file timestamp, or a machine-checked proof.

Theorem inventory (selection)

Claim	File	Content
R136	(framework)	directions declared in pairs $\{d, -d\}$
R138	PhaseRelationLocking.lean	phase relation locking
R143	MutualLocking.lean	interlock $\iff$ det $\neq 0$
R149	Pat4Phase.lean	four-phase interlock, S^3, $J^2 = -I$
R150	PatCountableInfinitPhaseUnification.lean	countable/continuum
R151	ContinuumPatGrid.lean	continuum = Pat-grid closure
R163	ContinuumConstructible.lean	endogenous, zero Nat
C001–C010	ZeroRelative/	heap, semiconjugacy, generation
C026–C038	ZeroRelative/	operator semantics, Q2_0, state structure